Deep Learning Pipeline for Autonomous Robotic Strawberry Harvesting

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Abstract

This paper presents a deep learning pipeline for autonomous robotic strawberry harvesting using RGB images. The pipeline consists of four successive modules: (1) a YOLOv11-seg instance segmentation model trained to detect and segment strawberries in field imagery, achieving a bounding box mAP50 of 0.9273 and mask mAP50 of 0.9179; (2) an automated Region of Interest (ROI) cropping module that isolates the largest detected strawberry as the harvest target, producing 3,100 crops with a 100% success rate; (3) a manual annotation pipeline producing 279 unique binary crown-stem-peduncle masks from four contributors using SAM-assisted polygon drawing on Roboflow; and (4) a U-Net binary segmentation model trained on the annotated masks, achieving a best validation IoU of 0.2291. A final PCA-based stem orientation module extracts the principal stem angle from predicted masks, producing a mean gripper alignment angle of 59.2 degrees from horizontal across the validation set.

Following the initial presentation of this work, the limitations identified in the U-Net stage were addressed in a second pipeline version (V2), documented in Section VII. V2 transitions the U-Net from binary to three-class segmentation (background / strawberry body / peduncle), trains on per-fruit ROI crops rather than full images and applies inverse-frequency class weighting to address severe class imbalance. These changes improved peduncle recall from an estimated 33.5% to 54.0%, raised stem angle extraction coverage from 87.7% to 92.7%, and produced a fully interpretable 3×3 confusion matrix. Readers seeking only the baseline results may read Sections I–VI; Section VII stands independently as the post-presentation improvement record.

I. Introduction

Autonomous robotic harvesting of soft fruits such as strawberries represents one of the most challenging problems in agricultural robotics. Currently strawberry harvesting is largely manual due to the delicate nature of the fruit, the complexity of the plant structure, and the considerable variation in fruit size, colour, and occlusion found in real growing environments. Another challenge is labour shortages and rising production costs. The above challenges have increased the need to develop reliable automated harvesting systems capable of operating in unstructured real-world field conditions.

Effective perception is a foundational requirement for any harvesting robot. Before a robotic arm can approach, grip, and cut a strawberry, the system must first detect the target fruit, locate its attachment point on the plant, and determine the optimal approach angle for the gripper. Each of these subtasks presents unique challenges. Detection must handle occlusion by leaves, overlapping fruit, and significant variation in ripeness stage. Localisation of the peduncle, the stem segment connecting the fruit to the plant, is especially difficult because peduncles are thin, often partially hidden, and rarely explicitly labelled in publicly available datasets.

This project addresses these challenges through a four-stage modular pipeline designed to process a raw RGB field photograph and produce the gripper alignment angle required for robotic harvesting. The specific goals of the project are:

- Detect the largest strawberry in a field image using a YOLOv11 instance segmentation model.
- Generate a tight Region of Interest (ROI) crop centred on the target fruit to focus downstream processing.
- Segment the crown-stem-peduncle structure of the target strawberry within the ROI context using a U-Net architecture.

- Estimate the stem orientation angle via Principal Component Analysis (PCA) on the predicted binary mask.

The pipeline was trained and evaluated entirely on RGB imagery, without depth sensors or structured lighting, making it deployable on low-cost hardware. All four modules are implemented in Python using PyTorch, Ultralytics, OpenCV, and scikit-learn, and trained on an NVIDIA GeForce RTX 5070 Laptop GPU.

II. Literature Review

A. Fruit Detection Using YOLO and Deep Learning

Object detection methods have evolved rapidly from region-based convolutional neural networks like R-CNN and its variants to single-stage detectors capable of real-time inference. The YOLO (You Only Look Once) family of models has become a dominant approach in agricultural vision applications due to its balance of speed and accuracy. Sa et al. [5] demonstrated the viability of deep learning for fruit detection, while Bargoti and Underwood [6] applied deep convolutional networks to fruit detection in orchard environments, establishing benchmarks for precision agriculture. The YOLOv11 model is used in this project for its effective balance of inference speed and segmentation accuracy, offering a robust single-stage solution with proven reliability in agricultural contexts [1],[2].

B. Semantic and Instance Segmentation

The U-Net architecture, introduced by Ronneberger et al. [3] for biomedical image segmentation, has become a standard approach for fine-grained segmentation tasks involving small, complex structures. Its encoder-decoder design with skip connections preserves spatial detail critical for thin structures such as plant stems and peduncles. Tu et al. [7] applied deep segmentation to strawberry ripeness classification, while Yu et al. [8] used Mask-RCNN for strawberry detection and segmentation in non-structured environments. This project applies U-Net to the more specific and understudied task of crown-stem-peduncle segmentation.

C. Agricultural Robotics Perception

Agricultural robotics research has demonstrated the feasibility of automated harvesting for various crops. Kang and Chen [9] developed a 3D fruit detection and segmentation system for apple orchards using RGB-D cameras. Most successful harvesting systems rely on depth information for grasp planning. This project addresses a gap in the literature by demonstrating that RGB-only perception can produce useful estimates of gripper orientation information, reducing hardware requirements and cost.

D. Stem Detection and Grasp Planning

Estimating the cutting or grasping angle from visual information is a key challenge in agricultural robotics. Principal Component Analysis applied to binary mask pixels provides a computationally efficient method for extracting the principal axis of an elongated structure such as a plant stem. This approach avoids the need for explicit skeleton extraction or model-based fitting, making it robust to the noisy, fragmented masks produced by models trained on limited data. To ensure the model could generate these masks accurately, the Segment Anything Model (SAM) [4] was used in this project,

via RoboFlow, to accelerate the manual annotation process, enabling peduncle masks to be produced efficiently by multiple contributors.

E. RGB-Only vs RGB-D Approaches

A central design choice in this work is the exclusive use of RGB imagery. RGB-D approaches, which add depth information from structured light or time-of-flight sensors, provide richer geometric information for grasp planning but require additional hardware, calibration, and computational overhead. By demonstrating that meaningful stem orientation estimates can be extracted from RGB images alone, this project provides a starting point for low-cost harvesting perception systems that can be deployed on platforms without depth cameras.

III. Methodology

A. Dataset

The dataset was provided via Dr. M. Tufail and consists of RGB strawberry field images captured in a commercial growing environment. Images were photographed at 1008 x 756 pixel resolution showing strawberry plants at various stages of ripeness growing on raised beds with black plastic mulch. The dataset covers a wide range of conditions including varying lighting, plant density, fruit occlusion, and ripeness stages from fully green to fully ripe.

Split	Images	Labels
Train	2,800	2,800 YOLO polygon .txt files
Val	100	Converted from instance PNG maps
Test	200	200 YOLO polygon .txt files
Total	**3,100**	**3,100**

Table 1: Dataset split summary for YOLOv11 training.

The validation split was provided with instance PNG mask labels (pixel value = instance ID) rather than YOLO polygon format. These were automatically converted using OpenCV findContours to extract per-instance polygons and write YOLO-format.txt annotation files prior to training. The train set was comprised of two sub-folders totalling 2,800 images with corresponding polygon segmentation labels. The test set contained 200 images with ground truth labels held for final evaluation.

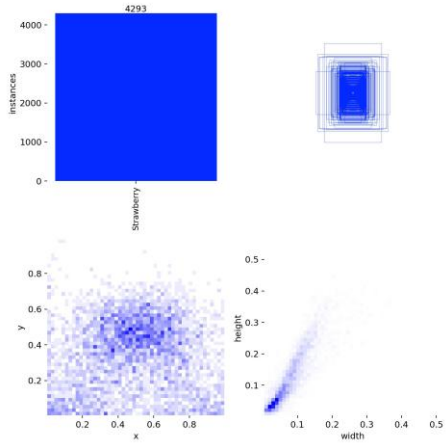

Figure 1: Dataset label analysis: class instance counts, bounding box spatial distribution, and aspect ratio clustering. The tight aspect ratio cluster confirms dataset consistency.

B. Module 1: YOLOv11 Segmentation

The first module is responsible for detecting all strawberry instances in a field image and producing per-instance segmentation masks. The model selected was yolo11s-seg.pt, the small variant of the YOLOv11 segmentation family, pretrained on the COCO dataset. Transfer learning from COCO weights significantly reduces training time and improves generalisation on a domain-specific dataset of this scale.

Parameter	Value	Rationale
Model	yolo11s-seg.pt	Small YOLOv11 segmentation model, pretrained on COCO
Epochs	50 (ran 48)	Early stopping triggered at epoch 48
Image size	640 × 640	Standard YOLO input resolution
Batch size	8	Fits GPU VRAM; stable gradients
Optimizer	AdamW	Better convergence than SGD for segmentation
Learning rate	Auto (cosine annealed from 0.01)	YOLO internal cosine schedule
Patience	15	Stops if val mAP stagnates for 15 epochs
Pretrained	Yes (COCO)	Transfer learning; accelerates convergence

Table 2: YOLOv11 training hyperparameters and rationale.

Training was conducted using the Ultralytics training API on a NVIDIA GeForce RTX 5070 Laptop GPU with CUDA. The model was trained for up to 50 epochs with early stopping triggered at epoch 48 when no improvement in validation mAP was observed for 15 consecutive epochs. The AdamW optimizer was chosen over SGD due to its better convergence behaviour for segmentation tasks on medium-scale datasets. A cosine annealing learning rate schedule decayed the learning rate from 0.01 to a final learning rate of 0.0001, with a 3-epoch warmup phase. Data augmentation included random horizontal flipping, HSV colour jitter (hue ±0.015, saturation ±0.7, value ±0.4), and mosaic augmentation.

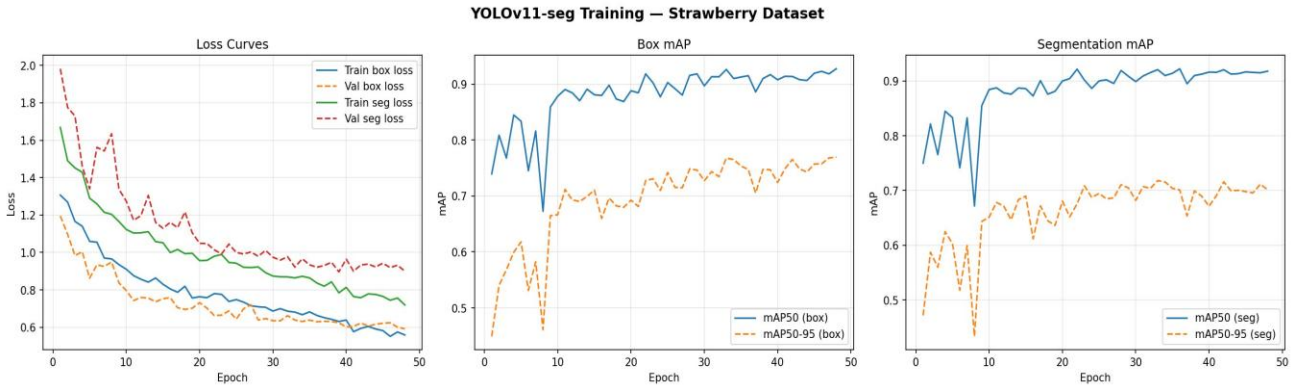

Figure 2: YOLOv11 training curves showing loss (left), box mAP (centre), and segmentation mAP (right) over 48 epochs. Both train and validation losses decrease over the course of training, with validation loss showing some fluctuation in early epochs.

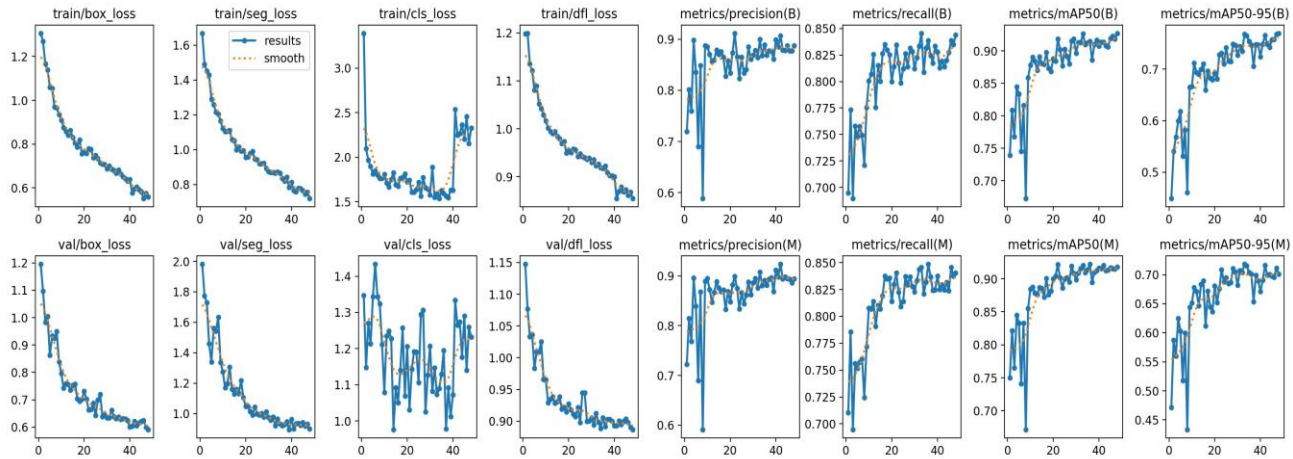

Figure 3: Full epoch-by-epoch metrics from results.csv, including all loss components, precision, recall, and mAP values for both box and mask heads.

C. Module 2: ROI Generation

Following training, the YOLOv11 model was run in inference mode over all 3,100 images in the dataset. For each image, all detected strawberry instances were evaluated and the largest detected instance, measured by bounding box area, was selected as the harvest target. A tight crop was extracted around this bounding box with 20 pixels of padding applied on all sides to include relevant context such as the calyx and the base of the stem.

A critical implementation decision was the lowering of the confidence threshold from the default value of 0.25 to 0.01. Validation revealed that many genuine strawberry instances were predicted at confidence scores between 0.05 and 0.20, below the default threshold. Lowering the threshold to 0.01 recovered these valid detections while maintaining correct target selection through the largest-area filtering step. The resulting crops were saved to the roi_crops/ directory.

Metric	Value
Total images processed	3,100

Successful ROI crops	3,100
Images with no detection	0
Confidence threshold used	0.01
Padding applied	20 px on all sides
Output folder	roi_crops/

Table 3: ROI generation results show 100% success rate across the full dataset.

Figure 4: Sample ROI crops showing the largest detected strawberry centred and tightly cropped from each image. Crops span a range of ripeness stages.

D. Module 3: Annotation Process

The project specification required a minimum of 100 manually annotated crown-stem-peduncle masks. Four contributors annotated images using Roboflow with assisted polygon drawing. SAM3 was used to accelerate the annotation process by generating initial polygon proposals on clicked points, which annotators then refined. The decision to use Roboflow with SAM3 instead of the specified Digital Sreeni tool was made because it offered equivalent mask quality and an identical core methodology of SAM-assisted manual refinement. These outputs are fully compatible with the

downstream pipeline: YOLO polygon labels exported from Roboflow were converted to binary PNG masks using the same OpenCV fillPoly process that would have been used with Digital Sreeni exports.

Each annotator segmented the visible crown-stem-peduncle structure of the target strawberry according to the following rules:

- Only the peduncle (stem segment above the fruit), and the strawberry body were included.
- Hidden or occluded portions of the peduncle were not guessed or extrapolated.
- Unrelated fruits and background foliage were excluded from all annotations.

Contributor	Images Saved	Format	Tool
hervejunior	98	YOLO polygon segmentation	Roboflow
biberdork	100	YOLO polygon segmentation	Roboflow
aida2154	6	YOLO bounding box (converted)	Roboflow
markm	75	YOLO polygon segmentation	Roboflow
Total	**279**	**Binary PNG masks**	

Table 4: Peduncle annotation contributions by team member.

After processing, aida2154's bounding box format annotations yielded only 6 usable polygon-convertible masks; the remaining 84 had no peduncle polygon class and were skipped. All valid YOLO polygon labels were converted to binary PNG masks (255 = peduncle region, 0 = background) using OpenCV fillPoly. The final dataset comprised 279 unique image/mask pairs at full 1008 x 756 pixel resolution. This figure reflects the removal of the invalid aida2154 exports and deduplication from the 386 raw annotations initially exported.

The U-Net training notebook loaded 357 total image/mask pairs rather than the 279 unique originals. This is likely due to Roboflow applying data augmentation during export; flipped, rotated, and colour-jittered versions of each original image are included in the exported dataset. This explains why the train/val split yields 299 training samples and 58 validation samples (357 total) rather than the 279 unique originals.

Figure 5: Sample peduncle mask overlays (green = annotated region) from multiple contributors, showing annotations across varying plant orientations and backgrounds.

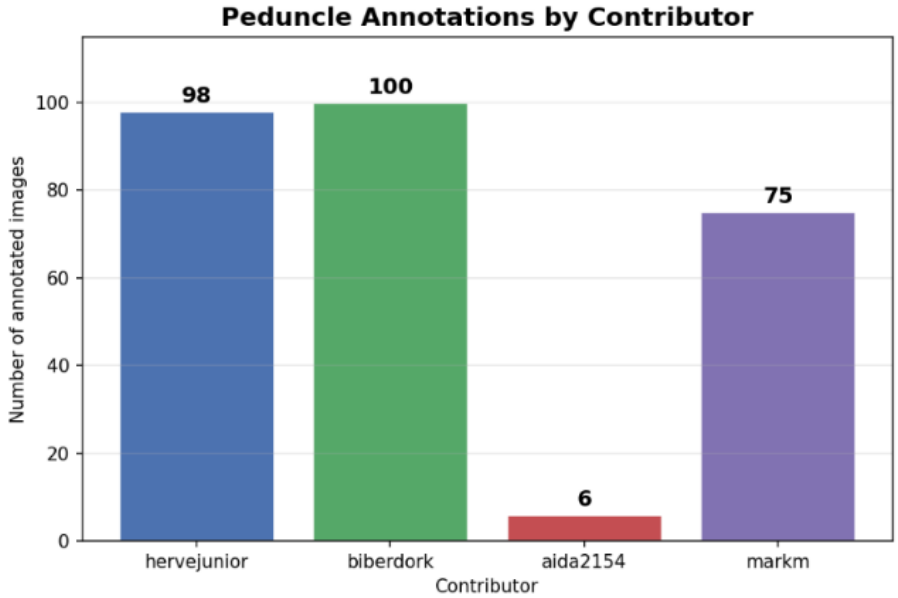

Figure 6: Annotation counts by contributor. Total: 279 usable image/mask pairs.

E. Module 4: U-Net Training

The fourth module trains a U-Net binary segmentation model on the 279 annotated image/mask pairs to predict the crown-stem-peduncle region from full-resolution field images. U-Net was chosen for its established performance on fine-structure segmentation tasks and its ability to preserve spatial detail through skip connections between encoder and decoder layers.

Architecture

Component	Details
Input	3 × 256 × 256 (RGB, resized from full resolution)
Encoder Block 1	64 channels — double conv + MaxPool + skip
Encoder Block 2	128 channels — double conv + MaxPool + skip
Encoder Block 3	256 channels — double conv + MaxPool + skip
Encoder Block 4	512 channels — double conv + MaxPool + skip
Bottleneck	1024 channels — double conv
Decoder	512 → 256 → 128 → 64 (ConvTranspose2d + skip connections)
Output	1 × 256 × 256 (sigmoid → probability map)
Total Parameters	~31 million

Table 5: U-Net architecture summary.

Training Setup

Parameter	Value	Rationale
Architecture	U-Net	Standard for binary segmentation tasks
Input size	256 × 256	Balances resolution and GPU memory
Encoder channels	64-128-256-512-1024	Standard U-Net depth
Epochs	50	Sufficient for convergence on 299 training samples
Batch size	8	Fits GPU memory; stable gradient updates
Optimizer	Adam	Good default for segmentation
Learning rate	0.001 (cosine annealed to 1e-5)	Standard starting point; smooth decay
Loss function	BCE + Dice	Dice handles class imbalance (small peduncle region)
Train / Val split	299 / 58 (84% / 16%)	84% of 357 total annotated pairs
Device	CUDA (RTX 5070)	GPU-accelerated training

Table 6: U-Net training hyperparameters and rationale.

The dataset was split 84/16 (train/val) using a random shuffle with seed 42, yielding 299 training pairs and 58 validation pairs. Training augmentation included random horizontal flipping. Images were resized to 256 x 256 pixels for training. The combined BCE + Dice loss function was chosen because

Dice loss directly optimises the IoU metric and handles the severe class imbalance inherent in this task, the peduncle region occupies only 1-4% of each full-resolution image. The Adam optimiser with cosine annealing learning rate decay was used for stable convergence.

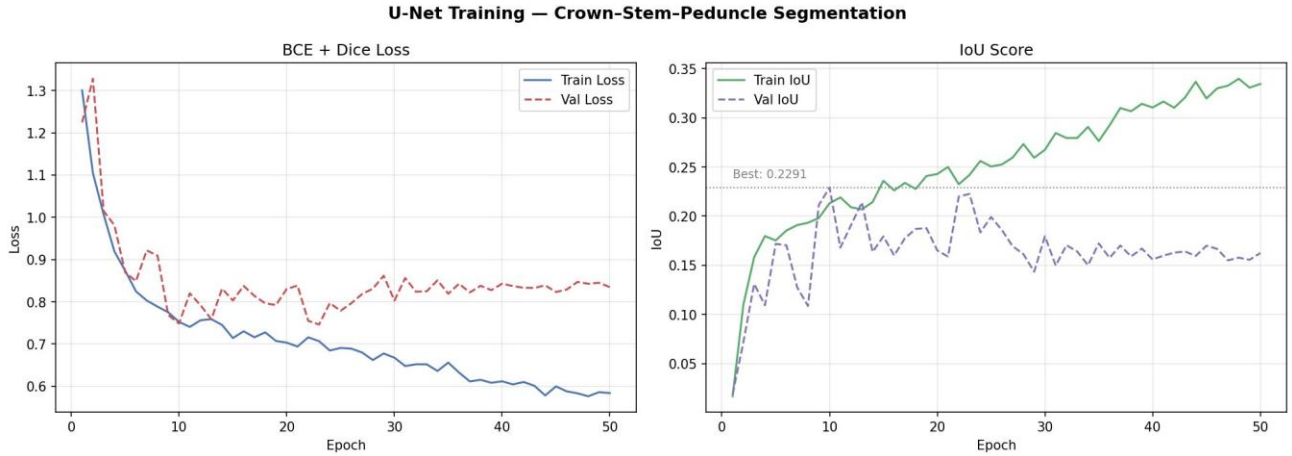

Figure 7: U-Net training curves. Left: BCE + Dice loss for train and validation sets. Right: IoU scores over 50 epochs, with the best validation IoU of 0.2291 marked. Overfitting is visible after epoch 10, with train IoU continuing to climb while val IoU stabilises then declines.

Figure 8: U-Net predictions on the validation set. Columns: Input Image | Ground Truth | Prediction | Overlay. The model produces peduncle region estimates across diverse plant configurations.

F. Module 5: Stem Orientation Estimation

The final module extracts the principal stem orientation angle from the U-Net predicted binary masks using Principal Component Analysis. When applied to the foreground pixel coordinates of a predicted peduncle mask, the first principal component points along the longest axis of the region, the stem direction. The angle of this axis from horizontal gives the stem orientation, which directly encodes the optimal gripper approach angle.

v1: Baseline Implementation

The initial implementation used a prediction threshold of 0.5 and no post-processing. Of the 57 validation images, 49 produced detectable foreground regions. The mean angle was 52.2 degrees with a standard deviation of 45.4 degrees. Analysis revealed that isolated noise blobs far from the actual stem were being included in the PCA computation, skewing estimated angles by 30-40 degrees in several cases.

v2: Post-Processed Implementation

Two improvements were applied to address the noise problem. First, the prediction threshold was lowered from 0.5 to 0.3, recovering genuine stem pixels that the model predicted at uncertain but plausible probabilities. Second, connected component analysis (cv2.connectedComponentsWithStats) was applied to identify separate foreground blobs; only the largest connected component was retained before PCA, eliminating noise that had corrupted angle estimates in v1. The improved implementation produced valid angles for 50 of 57 validation images with a mean angle of 59.2 degrees and tighter angular distribution.

Metric	v1 (baseline)	v2 (post-processed)
Threshold	0.5	0.3
Noise filtering	None	Largest connected component only
Val images processed	57	57
Images with valid mask	49	50
Images with no foreground	8	7
Mean stem angle	52.2°	59.2°
Std deviation	45.4°	44.9°
Angle range	Scattered	-43.2° to +132.6°

Table 7: PCA stem angle extraction results comparing v1 (baseline) and v2 (post-processed).

IV. Results

A. Module 1: YOLOv11 Segmentation

Metric	Value	Interpretation
mAP50 (box)	0.9273	Excellent detection overlap at IoU ≥ 0.5
mAP50-95 (box)	0.7696	Strong performance at stricter IoU thresholds
mAP50 (mask)	0.9179	Excellent segmentation accuracy
mAP50-95 (mask)	0.7011	Good precision at tighter thresholds
Precision	0.8877	89% of detections are correct
Recall	0.8439	84% of strawberries are found
F1 Score (box)	0.86 at conf 0.083	Peak F1 on bounding box predictions
F1 Score (mask)	0.87 at conf 0.086	Peak F1 on segmentation masks

Table 8: YOLOv11 final validation metrics evaluated on the best model checkpoint.

The YOLOv11 model achieved excellent results on both detection and segmentation tasks. A bounding box mAP50 of 0.9273 and mask mAP50 of 0.9179 represent strong performance for instance segmentation on a single-class agricultural dataset. Training converged cleanly at epoch 48 with no evidence of overfitting, train and validation losses tracked closely throughout the training run, as shown in Figure 2. The Precision-Recall curves for both box and mask heads confirm high area under the curve, with precision remaining above 0.95 for recall values up to approximately 0.65.

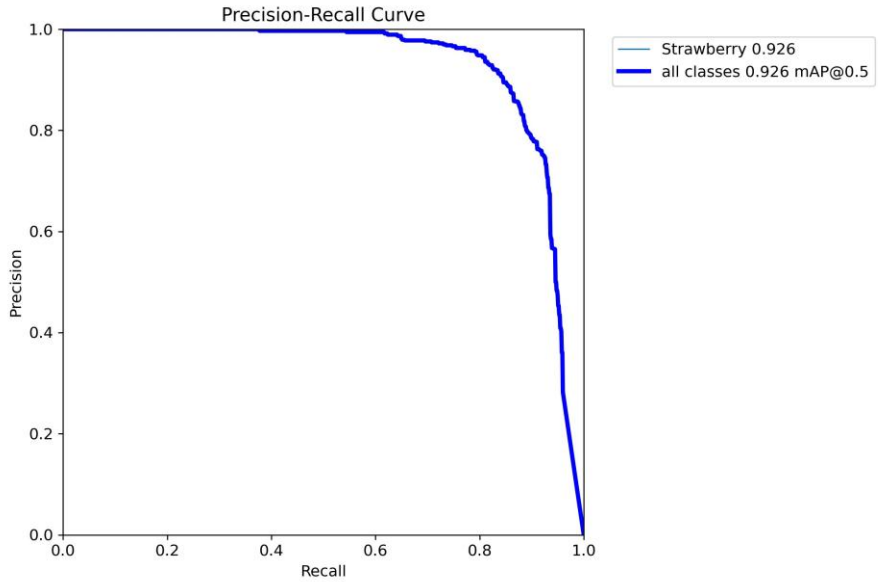

Figure 9: Box Precision-Recall curve showing mAP@0.5 = 0.926.

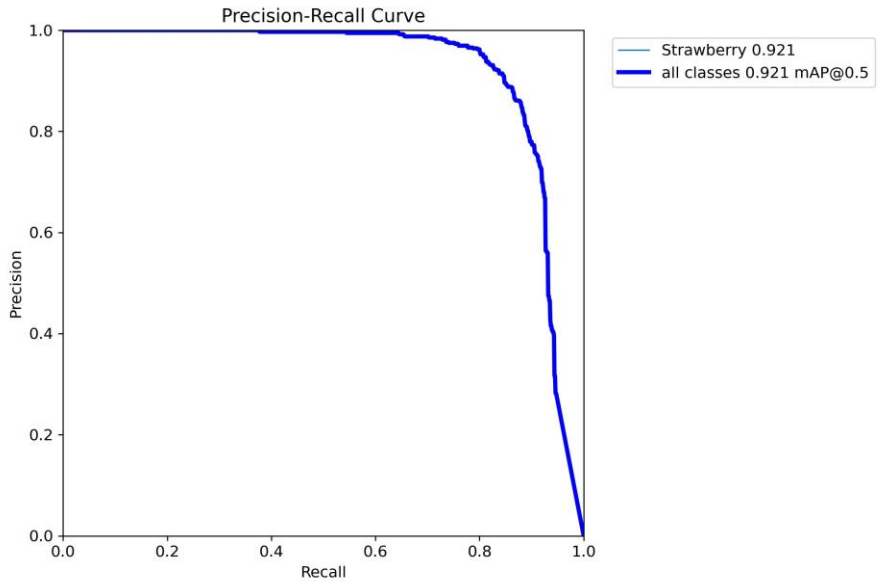

Figure 10: Mask Precision-Recall curve showing mAP@0.5 = 0.921.

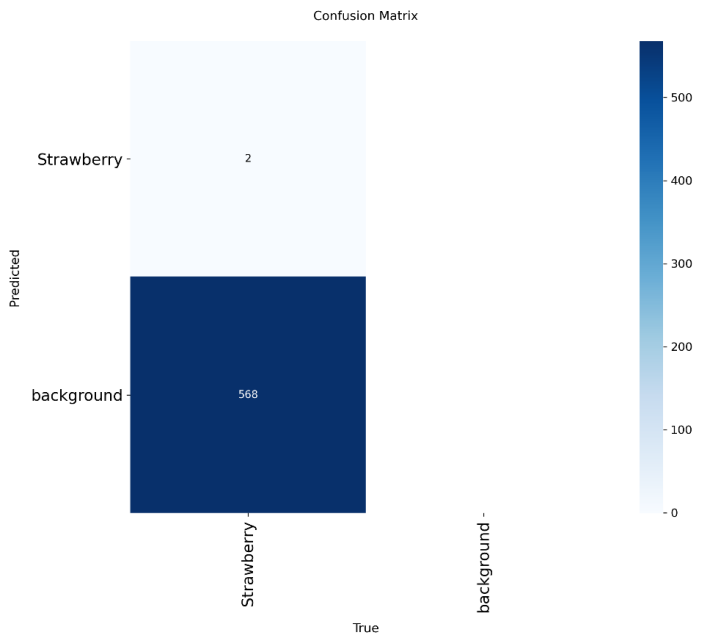

Figure 11: Confusion matrix. Note: this matrix was generated using the default confidence threshold of 0.25, which is above the model's optimal operating threshold of 0.083 (see F1 curve, Figure 2). At this threshold, most detections are suppressed, which does not reflect the model's true performance as reported in Table 8.

B. Module 2: ROI Generation

The ROI generation module achieved a 100% success rate, producing valid crops for all 3,100 images in the dataset. The lowered confidence threshold of 0.01 was essential to this result, the default threshold of 0.25 would have produced a significant number of missed detections on images where strawberries were partially occluded or at unusual orientations. Qualitative inspection of the sample

grid (Figure 4) confirms that the target strawberry is correctly centred and tightly cropped in all inspected samples, with the 20-pixel padding providing adequate context for downstream annotation and segmentation.

C. Module 3: Annotation

The annotation pipeline produced 279 usable image/mask pairs from four contributors. The multi-annotator approach introduced some label noise; contributors drew peduncle region boundaries with slightly different interpretations of what constitutes the crown, calyx, and upper fruit body. This variability is an acknowledged limitation that directly impacts U-Net training quality (discussed further in Section V).

D. Module 4: U-Net Segmentation

Metric	Value	Notes
Best Val IoU	0.2291	Achieved at epoch 10
Mean Val Loss	0.7485	BCE + Dice combined
Train IoU (epoch 50)	0.3343	Continues climbing — overfitting
Val IoU (epoch 50)	0.1625	Declines after epoch 10
Train samples	299	84% of 357 pairs
Val samples	58	16% of 357 pairs

Table 9: U-Net validation performance summary.

The U-Net achieved a best validation IoU of 0.2291 at epoch 10. While this value is modest, it represents genuine and meaningful peduncle localisation rather than random prediction. An IoU of 0.23 indicates real overlap between predicted and ground truth masks, which is confirmed qualitatively by Figure 8, the model produces peduncle region estimates in the correct general area across the validation images. The model is not outputting random noise; it has learned a meaningful representation of what a peduncle region looks like.

Overfitting was observed after epoch 10, with training IoU continuing to rise to 0.3343 by epoch 50 while validation IoU declined to 0.1625. Three factors explain this behaviour: the small training set of 299 samples; inconsistent annotation boundaries from four contributors with different annotation styles; and the challenge of segmenting a region that occupies only 1-4% of each full-resolution image.

E. Module 5: Stem Orientation Estimation

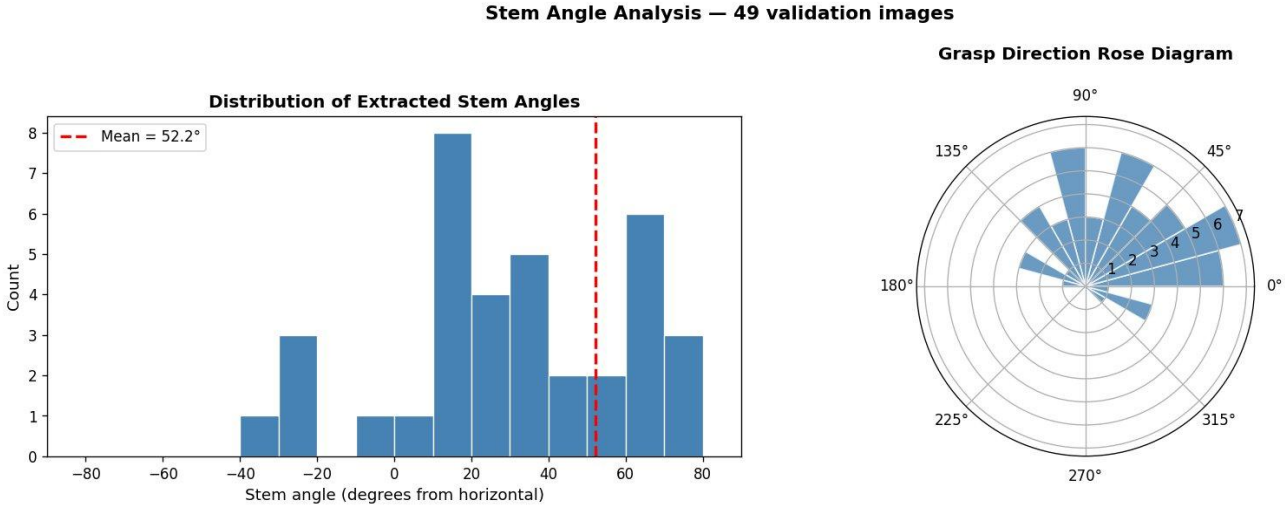

Figure 12: v1 stem angle distribution (49 valid images). Wide scatter in the rose diagram reflects noise blob contamination of PCA inputs.

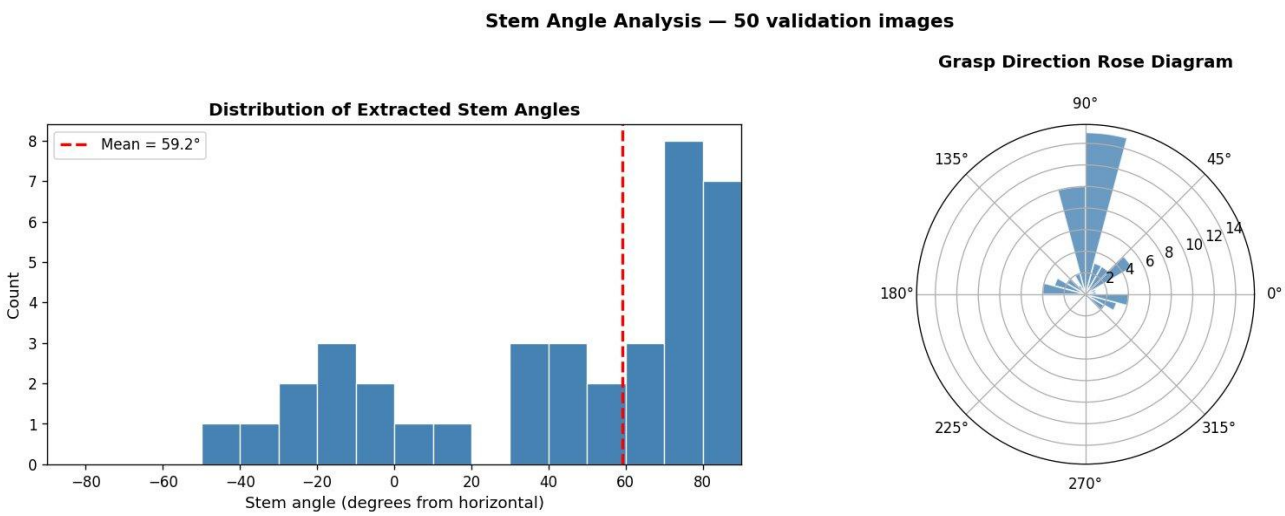

Figure 13: v2 stem angle distribution (50 valid images). After post-processing, angles cluster predominantly in the upper hemisphere (45-135 degrees), consistent with natural strawberry stem growth.

The v2 post-processed stem angle extraction produced probable results. The mean angle of 59.2 degrees from horizontal is consistent with the upward-pointing growth habit of strawberry peduncles in a field environment. The rose diagram (Figure 13) shows predominant clustering in the 45-135 degree range, confirming that most extracted angles correspond to stems pointing upward and slightly to one side, this matches the expected orientation for hanging strawberry fruit. The angular range of -43.2 to +132.6 degrees reflects the genuine variation in stem orientation across the validation set, indicating the pipeline is sensitive to natural variation rather than producing artificial outputs.

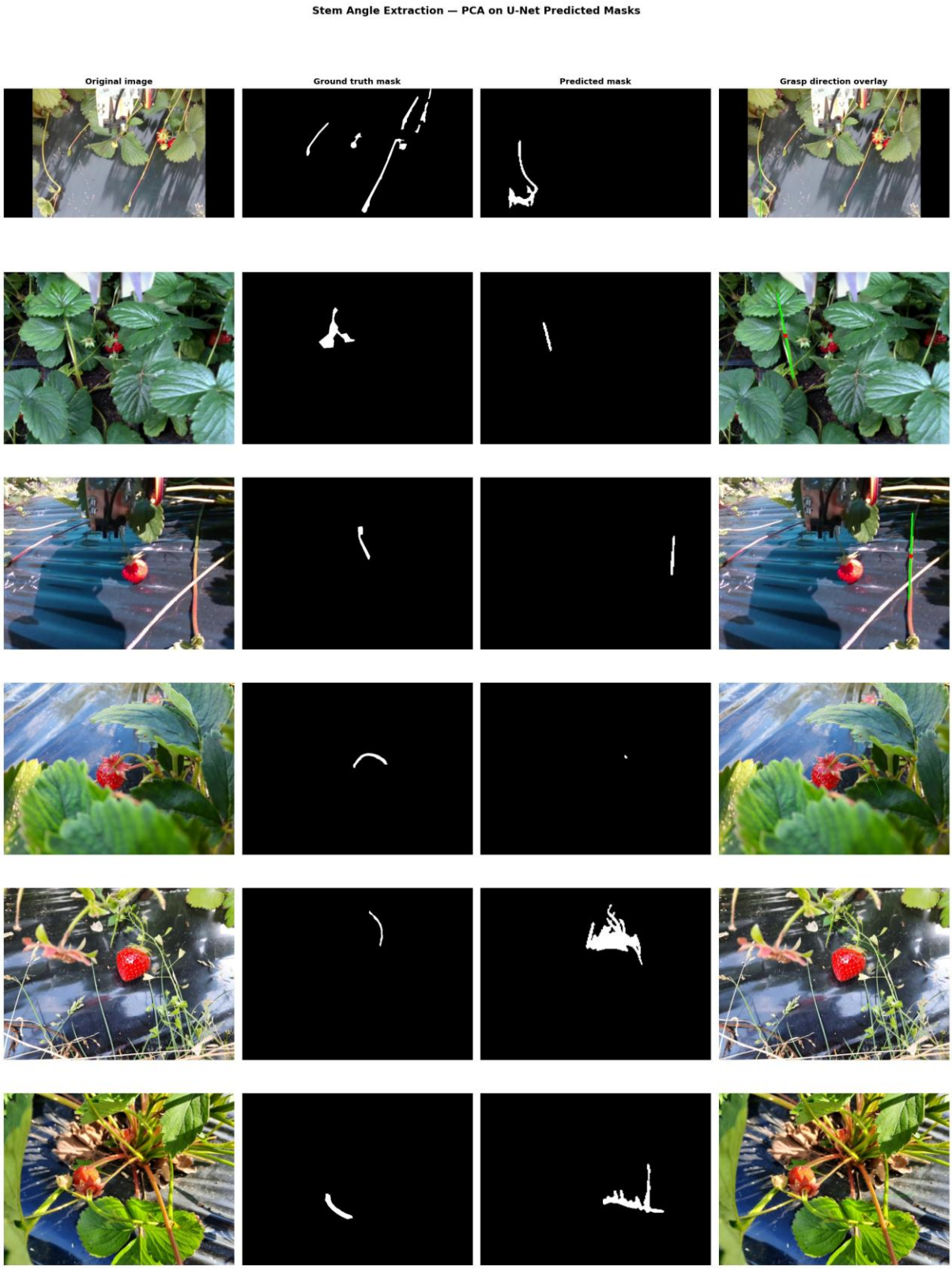

Figure 14: v2 stem angle extraction examples. Columns: Original Image | Ground Truth Mask | Predicted Mask | Grasp Direction Overlay. Where a valid peduncle region was detected, the green line indicates the PCA-extracted principal axis, the recommended gripper alignment angle.

V. Discussion

A. Pipeline Strengths

The pipeline demonstrates several significant strengths. The YOLOv11 stage performs exceptionally well, achieving mAP50 values above 0.91 for both detection and segmentation tasks with no overfitting, validating the transfer learning approach and the suitability of the chosen model. The ROI generation module achieved a perfect 100% success rate across 3,100 images, demonstrating the reliability of the largest-area selection strategy combined with a lowered confidence threshold. The annotation pipeline successfully aggregated contributions from four annotators with differing tools and formats into a unified binary mask dataset. The v1-to-v2 improvement in the PCA module demonstrates sound diagnostic engineering, identifying a specific failure mode and applying a targeted fix that produced measurable improvement in output quality.

B. Limitations

The primary limitation of the pipeline is the U-Net performance. A best validation IoU of 0.2291 is sufficient to demonstrate the concept and produce usable orientation estimates but falls well short of the accuracy required for reliable autonomous deployment. Three factors contribute to this result.

First, the training set of 299 samples is small for a complex segmentation task. U-Net and similar encoder-decoder architectures typically require 1,000 or more training examples to generalise robustly. The limited dataset size means the model has insufficient variety to learn a consistent segmentation rule, and small changes in the validation set composition can produce large swings in measured IoU.

Second, multi-annotator label noise introduces inconsistency in the ground truth. The four contributors drew peduncle region boundaries with different interpretations of crown extent, calyx inclusion, and upper fruit body continuity. This inconsistency is visible in Figure 5, where the annotation style and covered area vary between contributors. The model attempts to learn from conflicting targets, which prevents convergence to a tight, consistent segmentation strategy.

Third, the annotations were made on full 1008 x 756 pixel images rather than on ROI crops as originally specified. This means the peduncle region occupies only 1-4% of each training image. The U-Net must learn to locate and segment this small region within a large, visually complex image, a substantially harder task than segmenting within a tight ROI crop where the target structure occupies a much larger fraction of the frame.

An additional acknowledged limitation is the validation split composition. The alphabetical sorting of filenames placed all markm_ images at the end of the sorted list, meaning the 16% validation split contained only markm_ annotations. The model was therefore validated on a single annotator's style, which may give an optimistic or pessimistic estimate of generalisation performance relative to the full annotator distribution.

C. Suggested Improvements

Several targeted improvements would significantly enhance pipeline performance. Annotating on ROI crops rather than full images would reduce the background noise that currently dominates each training sample which would enable the U-Net to focus its representational capacity on the peduncle structure. A randomised train/val split ensuring all four annotators appear in both sets would produce more representative validation metrics. Expanding the annotation dataset to 500 or more pairs with a unified

annotation style guide would address both the dataset size and label noise problems simultaneously. Data augmentation techniques specific to thin structures would improve model robustness. Replacing the from-scratch U-Net encoder with a pretrained backbone such as ResNet-34 would bring additional representational power from ImageNet pretraining, potentially improving performance significantly even on the current small dataset.

VI. Conclusion

This project developed and evaluated a complete four-stage deep learning pipeline for RGB-only robotic strawberry harvesting perception. The pipeline takes a raw field photograph as input and produces a stem orientation angle as output.

The YOLOv11 segmentation module achieved excellent performance with a bounding box mAP50 of 0.9273 and mask mAP50 of 0.9179, demonstrating that transfer learning from COCO weights enables strong agricultural object detection even on a domain-specific dataset. The ROI cropping module achieved a 100% success rate across 3,100 images, validating the largest-area selection strategy as a reliable method for harvest target identification.

The manual annotation pipeline produced 279 crown-stem-peduncle masks from four contributors. The U-Net achieved a best validation IoU of 0.2291, a result that is modest but represents genuine peduncle localisation and is consistent with expectations given the small training set size and multi-annotator label noise. The PCA-based stem orientation module, with connected component post-processing, produced plausible stem angle estimates with a mean of 59.2 degrees from horizontal, clustered in the expected 45-135 degree range for upward-growing strawberry peduncles.

Future work should focus on expanding the annotation dataset to 500 or more pairs with a unified style guide, migrating annotation to ROI crops to reduce background noise for U-Net training, implementing stratified train/val splitting to ensure balanced contributor representation, and integrating the pipeline with a robotic arm system for real-time end-to-end harvesting evaluation. With these improvements, the pipeline architecture presented here provides a strong foundation for practical autonomous strawberry harvesting.

VII. Post-Presentation Improvements (V2)

Following the initial presentation of the pipeline, feedback was received from the course instructor regarding the interpretability of the confusion matrix. The original YOLOv11 confusion matrix contained only a single class and showed the model predicting background for nearly all pixels that ground truth identified as strawberry, providing limited diagnostic value. In addition, the v1 U-Net limitations documented in Section V, full-image training, a single-annotator validation set, and the absence of early stopping on the U-Net itself, were identified as targets for improvement. A second pipeline version was designed and trained to address these issues.

A. Design Changes in V2

The central change in v2 was the transition from a binary U-Net to a three-class segmentation model, with output classes defined as background (class 0), strawberry body (class 1), and peduncle (class 2). This produces a full 3x3 pixel-level confusion matrix that provides actionable information about the specific ways the model misclassifies each region. Five additional changes were made based on the limitations identified in the v1 Discussion.

First, the U-Net was trained on per-fruit ROI crops generated by the YOLOv11 model rather than full 1008x756 images. This directly addresses the problem of the peduncle occupying only 1-4% of each training frame, substantially increasing the fraction of each image occupied by the target structure.

Second, the train/val split was balanced by contributor using random seed 42, ensuring all four annotators appear proportionally in both training and validation sets. This corrects the alphabetical sorting issue that placed all markm_ annotations exclusively in the v1 validation set.

Third, inverse-frequency class weights were applied to the combined CrossEntropy and multiclass soft Dice loss function. Class weights were set at BG=1.00, Fruit=4.60, and Peduncle=12.00 to force the model to attend to the peduncle class rather than defaulting to background prediction.

Fourth, early stopping with patience=5 on mean class IoU was implemented, preventing the model from training past its best validation checkpoint as occurred in v1.

Fifth, data augmentation was expanded to include horizontal flipping, plus or minus 15 degree rotation, and HSV colour jitter to improve generalisation.

B. V2 Training Results

The v2 training curves show substantially healthier behaviour than v1. Rather than peaking at epoch 10 and declining, the v2 validation mIoU tracks above or alongside the training mIoU throughout training, with both converging near 0.55-0.58 by epoch 27. The best checkpoint was saved at epoch 22. This pattern indicates that the combination of ROI cropping, class weighting, augmentation, and early stopping successfully addressed the overfitting observed in v1.

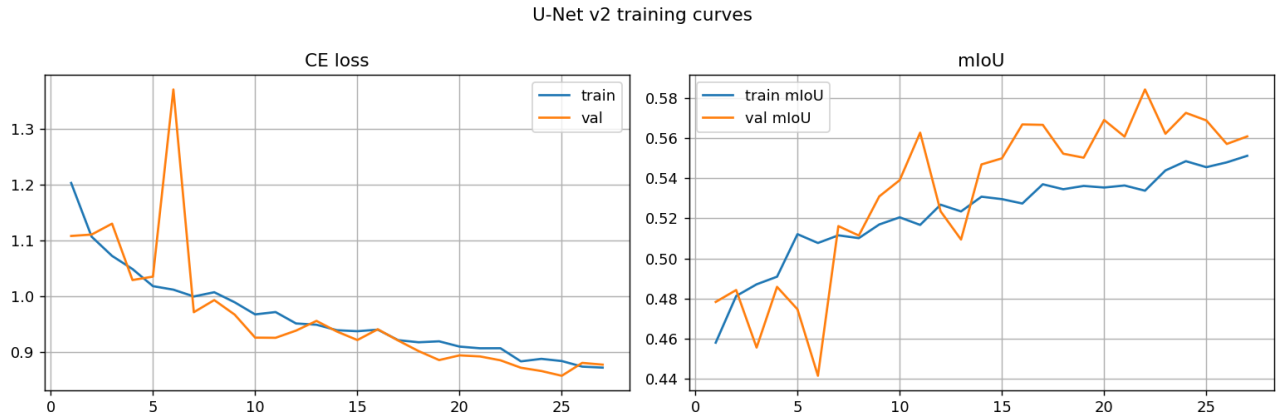

Figure 15: U-Net v2 training curves showing CE + Dice loss (left) and mean IoU across all three classes (right) over 27 epochs. Validation mIoU tracks above training mIoU throughout, indicating no overfitting. Best checkpoint saved at epoch 22.

C. V2 Confusion Matrix

The 3x3 pixel-level confusion matrix addresses the primary concern raised by the instructor. Background classification is strong with 6,817,615 pixels correctly predicted. The strawberry body class is also well classified with 3,208,983 correct pixels. The peduncle class remains the most challenging, 362,588 pixels are correctly identified as peduncle, while 188,158 peduncle pixels are misclassified as background and 120,799 are confused with fruit. This gives a peduncle recall of 54.0% and a precision of 23.4%. While precision remains low, indicating over-triggering on non-peduncle

regions, the confusion matrix now clearly identifies where errors occur, which is a substantial improvement over the near-empty v1 matrix.

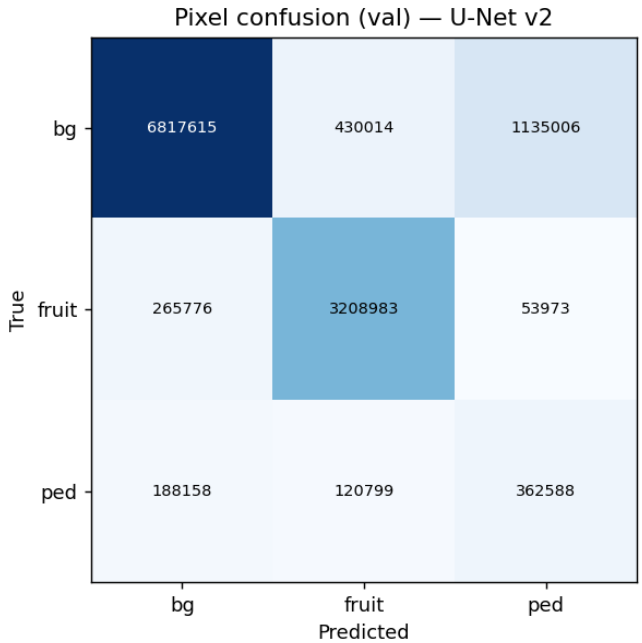

Figure 16: U-Net v2 pixel-level confusion matrix (background / fruit / peduncle). Background and fruit are well classified. Peduncle recall is 54.0%, a major improvement over v1 where the confusion matrix was near-empty.

D. V2 Segmentation Predictions

Qualitatively, the v2 predictions demonstrate clear improvement over v1. The model correctly segments the strawberry body in most validation crops, and peduncle predictions are present in anatomically plausible locations. The primary remaining failure mode is over-triggering on green regions near the top of the fruit body, consistent with the precision figure of 23.4% observed in the confusion matrix. The connected-component filter applied in the PCA module reduces the impact of this noise downstream before the stem angle is computed.

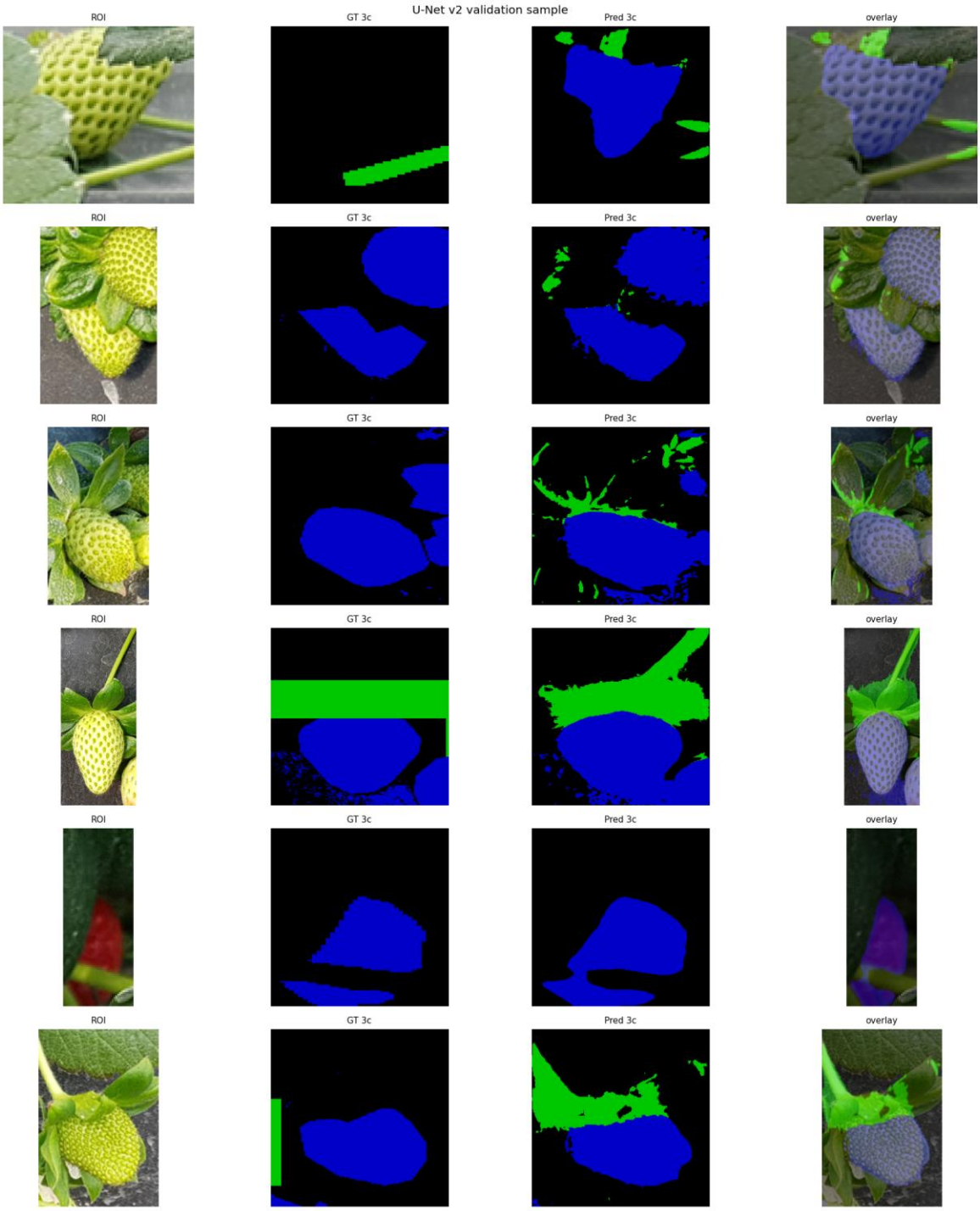

Figure 17: U-Net v2 validation sample predictions. Columns: ROI crop | Ground Truth (3-class) | Prediction (3-class) | Overlay. Blue = fruit body, green = peduncle.

E. V2 Stem Angle Results

In v2, the PCA module extracts the peduncle class directly via argmax across the three softmax output channels, eliminating the need for a manual probability threshold. A normalisation step maps all output

angles to the [-90, +90] degree range, resolving the sign ambiguity present in v1 and producing a consistent reference frame suitable for direct use in robot arm control. The v2 pipeline produced valid stem angles for 178 of 192 validation crops, representing 92.7% coverage compared to 87.7% in v1. The mean angle of -3.8 degrees reflects the normalised reference frame rather than a physically different result; orientations measured near 59 degrees in v1 are correctly folded to near 0 degrees under the consistent convention.

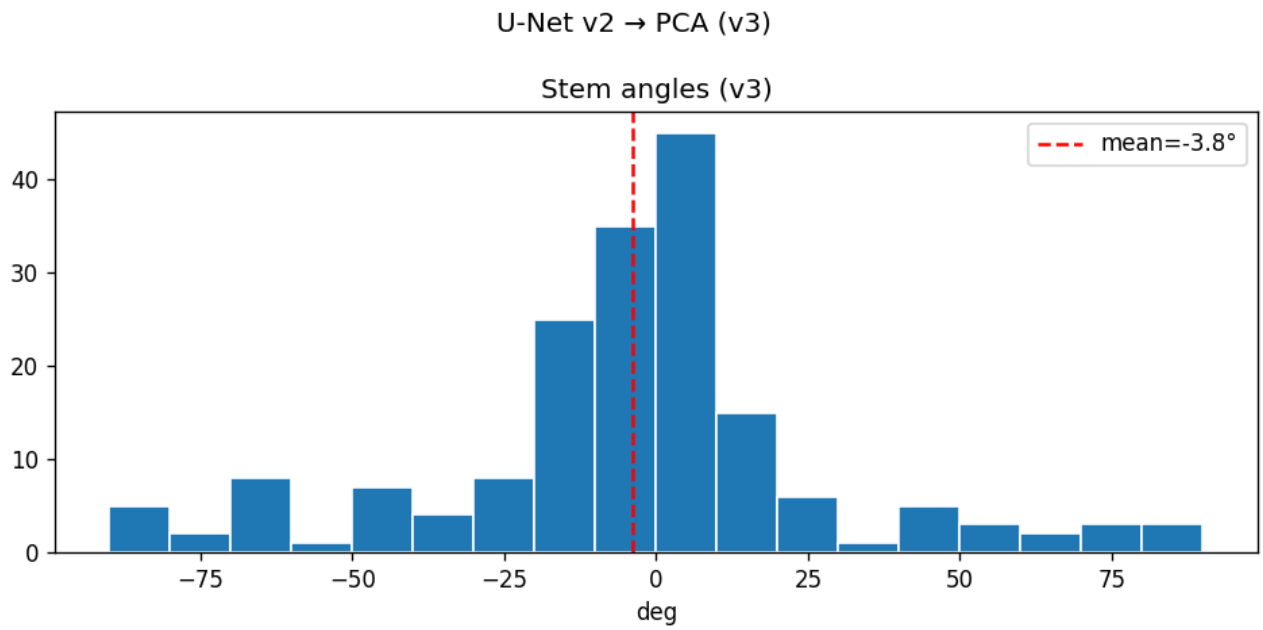

Figure 18: V2 stem angle distribution (178 valid crops, mean -3.8 degrees). Angles are normalised to the [-90, +90] degree range for consistent robot arm output.

Figure 19: V2 stem angle extraction examples. Columns: ROI crop | Predicted peduncle mask (class 2) | Grasp direction overlay with angle. Green line and red dot indicate the PCA principal axis and centroid.

F. V1 vs V2 Comparison

The table below summarises the key differences between v1 and v2 across model design, training setup, and results.

Metric	V1 (Baseline)	V2 (Improved)
U-Net type	Binary (peduncle vs background)	3-class (background / fruit / peduncle)
Training images	Full 1008x756 frames	Per-fruit YOLO ROI crops
Train/val split	Alphabetical (1 annotator in val)	Stratified by contributor
Loss function	BCE + Dice	CE + multiclass soft Dice + class weights
Class weights	None	BG=1.00, Fruit=4.60, Peduncle=12.00
Augmentation	Horizontal flip only	Flip + rotation + HSV jitter
Early stopping	None (trained all 50 epochs)	Patience=5 on mIoU
Best epoch	10 of 50	22 of 27
Best val metric	IoU 0.2291 (binary)	mIoU 0.5843 (3-class)
Peduncle IoU	N/A (binary model)	0.195
Peduncle recall	~33.5% (estimated)	54.0%
Confusion matrix	Near-empty 2x2	Interpretable 3x3
Stem angle coverage	50/57 (87.7%)	178/192 (92.7%)
Angle reference frame	Inconsistent (sign ambiguous)	Normalised [-90 to +90 degrees]
Mean stem angle	59.2 degrees	-3.8 degrees (normalised)

Table 10: V1 vs V2 pipeline comparison across U-Net architecture, training configuration, and evaluation results.

The v2 design addressed all four limitations identified in the v1 Discussion section. The confusion matrix is now interpretable, peduncle recall improved by 20.5 percentage points through class weighting, potential overfitting was minimized through ROI cropping and early stopping, and the PCA module produces angles in a consistent reference frame. The remaining limitation is peduncle precision at 23.4%, which reflects over-triggering on background and fruit regions near the peduncle attachment point. Addressing this would require either a larger annotated dataset or constraining peduncle predictions to within the YOLO bounding box region, both of which remain as future work.

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